

rdi_validated_ensemble

scalars_validated_ensemble_rdi

key	value
skipped_empty_gt	0
main_kwargs_applied.substitute	True
main_kwargs_applied.normalize_final_letters	True
step2_stats.n_models	9
step2_stats.spearman	0.9833
step2_stats.kendall_b	0.9444
step2_stats.spearman_exact_p_onetailed	2.4802e-05
step2_stats.kendall_exact_p_onetailed	2.4802e-05
step2_stats.p_method	exact
step2_stats.n_permutations	362880
step2_stats.min_attainable_p	2.7557e-06
step2_stats.pearson	0.997
step2_stats.pearson_p	4.9389e-09
step2_stats.mean_wer_compression	6.5626
step3_determinism.identical_outputs	3289
step3_determinism.n	3289
step3_determinism.deterministic	True
step3_determinism.main_kwargs.substitute	True
step3_determinism.main_kwargs.normalize_final_letters	True
step4_resolution_true.reference	true
step4_resolution_true.n_boot	10000
step4_resolution_true.n_samples	3290
step4_resolution_true.min_resolved_gap_wer	2.5972
step4_resolution_true.max_unresolved_gap_wer	0.3437
step4_resolution_true.n_resolved	34
step4_resolution_true.n_pairs	36
step4_resolution_pseudo.reference	pseudo
step4_resolution_pseudo.n_boot	10000
step4_resolution_pseudo.n_samples	3289
step4_resolution_pseudo.min_resolved_gap_wer	0.7488
step4_resolution_pseudo.max_unresolved_gap_wer	0.3942
step4_resolution_pseudo.n_resolved	35
step4_resolution_pseudo.n_pairs	36
step8_permutation_summary.canonical_order	seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb+cohere
step8_permutation_summary.n_orders	6

key	value
step8_permutation_summary.n_transcripts	3289
step8_permutation_summary.total_differing	0
step8_permutation_summary.invariant	True
step8_permutation_summary.main_kwargs.substitute	True
step8_permutation_summary.main_kwargs.normalize_final_letters	True

step0_cross_wer_validated_ensemble_rdi

	seamless_m4t	whisper_v3	whisper_v2	fastconformer	xlsr	w2vbert	hubert	wav2vec_sb	cohere
seamless_m4t	0	29.39	31.86	29.99	80.69	69.86	54.5	83.85	26.72
whisper_v3	29.39	0	21.85	26.04	77.81	67.27	50.1	81.93	24.1
whisper_v2	31.86	21.85	0	28.73	77.92	67.69	50.6	81.91	27.04
fastconformer	29.99	26.04	28.73	0	77.97	67.4	50.2	81.85	24.25
xlsr	80.69	77.81	77.92	77.97	0	84.09	84.13	81	90.5
w2vbert	69.86	67.27	67.69	67.4	84.09	0	75.41	86.68	77.97
hubert	54.5	50.1	50.6	50.2	84.13	75.41	0	80.85	54.19
wav2vec_sb	83.85	81.93	81.91	81.85	81	86.68	80.85	0	98.91
cohere	26.72	24.1	27.04	24.25	90.5	77.97	54.19	98.91	0

step0_err_corr_validated_ensemble_rdi

	seamless_m4t	whisper_v3	whisper_v2	fastconformer	xlsr	w2vbert	hubert	wav2vec_sb	cohere
seamless_m4t	1	0.8006	0.7833	0.8142	0.5906	0.63	0.6806	0.5712	0.7823
whisper_v3	0.8006	1	0.8611	0.8286	0.6148	0.6497	0.7224	0.5906	0.7771
whisper_v2	0.7833	0.8611	1	0.8089	0.6331	0.6713	0.7269	0.6097	0.7474
fastconformer	0.8142	0.8286	0.8089	1	0.6318	0.6726	0.7371	0.6122	0.7804
xlsr	0.5906	0.6148	0.6331	0.6318	1	0.8754	0.8139	0.9353	0.5115
w2vbert	0.63	0.6497	0.6713	0.6726	0.8754	1	0.8289	0.8727	0.5505
hubert	0.6806	0.7224	0.7269	0.7371	0.8139	0.8289	1	0.7939	0.6018
wav2vec_sb	0.5712	0.5906	0.6097	0.6122	0.9353	0.8727	0.7939	1	0.4957
cohere	0.7823	0.7771	0.7474	0.7804	0.5115	0.5505	0.6018	0.4957	1

step0_flags_validated_ensemble_rdi

pair	cross_wer	err_corr	note
seamless_m4t whisper_v3	29.39	0.8006	unusually similar hypotheses - suspect shared lab/data/teacher
seamless_m4t whisper_v2	31.86	0.7833	unusually similar hypotheses - suspect shared lab/data/teacher
seamless_m4t fastconformer	29.99	0.8142	unusually similar hypotheses - suspect shared lab/data/teacher

pair	cross_wer	err_corr	note
seamless_m4t cohere	26.72	0.7823	unusually similar hypotheses - suspect shared lab/data/teacher
whisper_v3 whisper_v2	21.85	0.8611	unusually similar hypotheses - suspect shared lab/data/teacher
whisper_v3 fastconformer	26.04	0.8286	unusually similar hypotheses - suspect shared lab/data/teacher
whisper_v3 cohere	24.1	0.7771	unusually similar hypotheses - suspect shared lab/data/teacher
whisper_v2 fastconformer	28.73	0.8089	unusually similar hypotheses - suspect shared lab/data/teacher
whisper_v2 cohere	27.04	0.7474	unusually similar hypotheses - suspect shared lab/data/teacher
fastconformer cohere	24.25	0.7804	unusually similar hypotheses - suspect shared lab/data/teacher

step1_bias_validated_ensemble_rdi

model	role	n_samples	spearman_full	spearman_loo	kendall_full	kendall_loo	delta_spearman	delta_kendall
cohere	voter	3289	0.4367	0.4767	0.329	0.3557	0.04	0.0266
seamless_m4t	voter	3289	0.5708	0.6112	0.435	0.4641	0.0404	0.0291
whisper_v3	voter	3289	0.5824	0.6316	0.4431	0.4791	0.0492	0.0361
fastconformer	voter	3289	0.592	0.6375	0.4509	0.4851	0.0454	0.0343
whisper_v2	ranked_only(control)	3289	0.651	0.651	0.4987	0.4987	0	0
hubert	voter	3289	0.8159	0.831	0.6567	0.6754	0.015	0.0187
wav2vec_sb	voter	3289	0.8634	0.8689	0.7549	0.761	0.0056	0.0061
xlsr	ranked_only(control)	3289	0.8706	0.8706	0.7507	0.7507	0	0
w2vbert	voter	3289	0.8713	0.8774	0.7342	0.7423	0.0061	0.0081

step2_ranking_validated_ensemble_rdi

model	role	n_samples	wer_true	cer_true	wer_pseudo_loo	cer_pseudo_loo	rank_true	rank_pseudo	wer_compression
cohere	voter	3289	26.4854	14.1468	20.4852	12.3275	1	1	6.0002
fastconformer	voter	3289	31.9281	14.799	21.9683	10.9025	2	3	9.9598
whisper_v3	voter	3289	32.2718	14.4793	21.574	10.2918	3	2	10.6978
whisper_v2	ranked_only	3289	34.8691	15.8537	22.7171	10.9808	4	4	12.152
seamless_m4t	voter	3289	34.912	17.7268	25.6018	13.9828	5	5	9.3102
hubert	voter	3289	53.6297	22.7558	48.5705	19.8795	6	6	5.0591
w2vbert	voter	3289	69.022	28.3899	66.0015	26.2696	7	7	3.0205
xlsr	ranked_only	3289	78.6661	35.7994	76.9979	34.496	8	8	1.6681
wav2vec_sb	voter	3289	82.3996	41.7731	81.2044	40.4524	9	9	1.1952

step3_model_dropout_validated_ensemble_rdi

model	times_filtered_out	times_in_pool	filtered_out_rate
wav2vec_sb	21869	23023	0.9499

model	times_filtered_out	times_in_pool	filtered_out_rate
w2vbert	13492	23023	0.586
hubert	5680	23023	0.2467
seamless_m4t	694	23023	0.0301
whisper_v3	545	23023	0.0237
cohere	525	23023	0.0228
fastconformer	459	23023	0.0199

step3_runs_validated_ensemble_rdi

run	n_voters	n_samples	filter_fire_rate	tie_rate	refsel_most_common_words	refsel_longest_match_validation
full_pool	7	3289	0.9976	0.1591	0.7239	0.2761
LOO_minus_seamless_m4t	6	3289	0.9815	0.1655	0.6838	0.3162
LOO_minus_whisper_v3	6	3289	0.9818	0.1645	0.657	0.343
LOO_minus_fastconformer	6	3289	0.9821	0.163	0.6695	0.3305
LOO_minus_w2vbert	6	3289	0.9915	0.1145	0.6671	0.3329
LOO_minus_hubert	6	3289	0.9957	0.1272	0.6774	0.3226
LOO_minus_wav2vec_sb	6	3289	0.9894	0.1141	0.6589	0.3411
LOO_minus_cohere	6	3289	0.9836	0.1796	0.7066	0.2934

step4_pairs_pseudo_validated_ensemble_rdi

pair	adjacent	gap_wer	preserved_frac	resolved_95
whisper_v3 < fastconformer	True	0.3942	0.9418	False
fastconformer < whisper_v2	True	0.7488	0.9928	True
cohere < whisper_v3	True	1.0888	1	True
whisper_v3 < whisper_v2	False	1.1431	1	True
cohere < fastconformer	False	1.4831	1	True
cohere < whisper_v2	False	2.2319	1	True
whisper_v2 < seamless_m4t	True	2.8847	1	True
fastconformer < seamless_m4t	False	3.6335	1	True
whisper_v3 < seamless_m4t	False	4.0278	1	True
xlsr < wav2vec_sb	True	4.2064	1	True
cohere < seamless_m4t	False	5.1166	1	True
w2vbert < xlsr	True	10.9965	1	True
w2vbert < wav2vec_sb	False	15.2029	1	True
hubert < w2vbert	True	17.431	1	True
seamless_m4t < hubert	True	22.9687	1	True
whisper_v2 < hubert	False	25.8535	1	True
fastconformer < hubert	False	26.6023	1	True

pair	adjacent	gap_wer	preserved_frac	resolved_95
whisper_v3 < hubert	False	26.9965	1	True
cohere < hubert	False	28.0854	1	True
hubert < xlsr	False	28.4274	1	True
hubert < wav2vec_sb	False	32.6338	1	True
seamless_m4t < w2vbert	False	40.3997	1	True
whisper_v2 < w2vbert	False	43.2844	1	True
fastconformer < w2vbert	False	44.0332	1	True
whisper_v3 < w2vbert	False	44.4275	1	True
cohere < w2vbert	False	45.5163	1	True
seamless_m4t < xlsr	False	51.3961	1	True
whisper_v2 < xlsr	False	54.2809	1	True
fastconformer < xlsr	False	55.0297	1	True
whisper_v3 < xlsr	False	55.4239	1	True
seamless_m4t < wav2vec_sb	False	55.6026	1	True
cohere < xlsr	False	56.5128	1	True
whisper_v2 < wav2vec_sb	False	58.4873	1	True
fastconformer < wav2vec_sb	False	59.2361	1	True
whisper_v3 < wav2vec_sb	False	59.6304	1	True
cohere < wav2vec_sb	False	60.7192	1	True

step4_pairs_true_validated_ensemble_rdi

pair	adjacent	gap_wer	preserved_frac	resolved_95
whisper_v2 < seamless_m4t	True	0.043	0.5154	False
fastconformer < whisper_v3	True	0.3437	0.8962	False
whisper_v3 < whisper_v2	True	2.5972	1	True
whisper_v3 < seamless_m4t	False	2.6401	1	True
fastconformer < whisper_v2	False	2.9409	1	True
fastconformer < seamless_m4t	False	2.9839	1	True
xlsr < wav2vec_sb	True	3.7334	1	True
cohere < fastconformer	True	5.4426	1	True
cohere < whisper_v3	False	5.7863	1	True
cohere < whisper_v2	False	8.3835	1	True
cohere < seamless_m4t	False	8.4264	1	True
w2vbert < xlsr	True	9.6438	1	True
w2vbert < wav2vec_sb	False	13.3773	1	True
hubert < w2vbert	True	15.392	1	True
seamless_m4t < hubert	True	18.7172	1	True
whisper_v2 < hubert	False	18.7601	1	True

pair	adjacent	gap_wer	preserved_frac	resolved_95
whisper_v3 < hubert	False	21.3573	1	True
fastconformer < hubert	False	21.701	1	True
hubert < xlsr	False	25.0358	1	True
cohere < hubert	False	27.1436	1	True
hubert < wav2vec_sb	False	28.7692	1	True
seamless_m4t < w2vbert	False	34.1091	1	True
whisper_v2 < w2vbert	False	34.1521	1	True
whisper_v3 < w2vbert	False	36.7493	1	True
fastconformer < w2vbert	False	37.093	1	True
cohere < w2vbert	False	42.5356	1	True
seamless_m4t < xlsr	False	43.753	1	True
whisper_v2 < xlsr	False	43.796	1	True
whisper_v3 < xlsr	False	46.3931	1	True
fastconformer < xlsr	False	46.7368	1	True
seamless_m4t < wav2vec_sb	False	47.4864	1	True
whisper_v2 < wav2vec_sb	False	47.5294	1	True
whisper_v3 < wav2vec_sb	False	50.1265	1	True
fastconformer < wav2vec_sb	False	50.4703	1	True
cohere < xlsr	False	52.1794	1	True
cohere < wav2vec_sb	False	55.9128	1	True

step5_baselines_validated_ensemble_rdi

reference	n_models	spearman	kendall_b	spearman_exact_p_onetailed	kendall_exact_p_onetailed	p_method	n_permutations	min_attainable_p	mean_abs_wer_err
LOO ensemble (proposed)	9	0.9833	0.9444	2.4802e-05	2.4802e-05	exact	362880	2.7557e-06	6.5626
full ensemble (biased)	9	0.9667	0.8889	8.2672e-05	0.0001	exact	362880	2.7557e-06	9.4589
single model: seamless_m4t	8	0.9762	0.9286	0.0002	0.0002	exact	40320	2.4802e-05	1.6551
LOO ensemble matched to n-1 (vs seamless_m4t)	8	0.9762	0.9286	0.0002	0.0002	exact	40320	2.4802e-05	6.2191
single model: whisper_v3	8	0.9286	0.8571	0.0011	0.0009	exact	40320	2.4802e-05	4.1563
LOO ensemble matched to n-1 (vs whisper_v3)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	6.0457
single model: fastconformer	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	3.0881
LOO ensemble matched to n-1 (vs fastconformer)	8	1	1	2.4802e-05	2.4802e-05	exact	40320	2.4802e-05	6.1379
single model: w2vbert	8	0.5	0.4286	0.1081	0.0894	exact	40320	2.4802e-05	32.2269
LOO ensemble matched to n-1 (vs w2vbert)	8	0.9762	0.9286	0.0002	0.0002	exact	40320	2.4802e-05	7.0053
single model: hubert	8	0.8571	0.7857	0.0054	0.0028	exact	40320	2.4802e-05	13.9307
LOO ensemble matched to n-1 (vs hubert)	8	0.9762	0.9286	0.0002	0.0002	exact	40320	2.4802e-05	6.7505
single model: wav2vec_sb	8	-0.6667	-0.5	0.9653	0.9695	exact	40320	2.4802e-05	49.6633
LOO ensemble matched to n-1 (vs wav2vec_sb)	8	0.9762	0.9286	0.0002	0.0002	exact	40320	2.4802e-05	7.2335
single model: cohere	8	0.9524	0.8571	0.0006	0.0009	exact	40320	2.4802e-05	4.4017

reference	n_models	spearman	kendall_b	spearman_exact_p_onetailed	kendall_exact_p_onetailed	p_method	n_permutations	min_attainable_p	mean_abs_wer_err
LOO ensemble matched to n-1 (vs cohere)	8	0.9762	0.9286	0.0002	0.0002	exact	40320	2.4802e-05	6.6329

step6_contribution_validated_ensemble_rdi

model_removed	delta_ensemble_wer	delta_ranking_spearman	helps_reference_quality	helps_ranking	delta_pairs_wrong	delta_pairs_wrong_resolvable
seamless_m4t	1.0408	-0.15	True	False	5	5
whisper_v3	0.5013	-0.1167	True	False	3	3
fastconformer	0.623	-0.1	True	False	3	4
w2vbert	-0.1074	0	False	True	0	0
hubert	-0.1934	0	False	True	0	0
wav2vec_sb	-0.1599	0	False	True	0	0
cohere	2.7524	-0.05	True	False	2	2

step6_pools_validated_ensemble_rdi

pool	k	ensemble_wer_vs_true	ensemble_cer_vs_true	ranking_spearman	ranking_exact_p	ranking_kendall_b	ranking_kendall_exact_p	n_pairs_wrong	n_pairs_wrong_resolvable	worst_wrong_gap_wer	worst_wrong_pair
seamless_m4t+whisper_v3+fastconformer+w2vbert+wav2vec_sb+cohere	6	25.8862	12.1146	0.9833	2.4802e-05	0.9444	2.4802e-05	1	0	0	
seamless_m4t+whisper_v3+fastconformer+w2vbert+cohere	5	25.9125	12.1303	0.9833	2.4802e-05	0.9444	2.4802e-05	1	0	0	
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb+cohere	5	25.9173	12.137	0.9833	2.4802e-05	0.9444	2.4802e-05	1	0	0	
seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert+cohere	6	25.9196	12.2337	0.9833	2.4802e-05	0.9444	2.4802e-05	1	0	0	
seamless_m4t+whisper_v3+fastconformer+hubert+cohere	5	25.965	12.1713	0.9833	2.4802e-05	0.9444	2.4802e-05	1	0	0	
seamless_m4t+whisper_v3+fastconformer+hubert+wav2vec_sb+cohere	6	25.9722	12.3352	0.9833	2.4802e-05	0.9444	2.4802e-05	1	0	0	
seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb+cohere	7	26.0796	12.3895	0.9833	2.4802e-05	0.9444	2.4802e-05	1	0	0	
seamless_m4t+whisper_v3+fastconformer+cohere	4	26.0885	12.1473	0.9833	2.4802e-05	0.9444	2.4802e-05	1	0	0	
seamless_m4t+whisper_v3+wav2vec_sb+cohere	4	26.4967	12.4817	0.9167	0.0007	0.8333	0.0004	3	3	8.3835	cohere whisper_v2
seamless_m4t+fastconformer+hubert+wav2vec_sb+cohere	5	26.4997	12.581	0.95	0.0002	0.8889	0.0001	2	1	5.7911	cohere whisper_v3
seamless_m4t+whisper_v3+hubert+wav2vec_sb+cohere	5	26.5522	12.4986	0.9167	0.0007	0.8333	0.0004	3	3	8.3837	cohere whisper_v2
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb+cohere	5	26.569	12.52	0.9167	0.0007	0.8333	0.0004	3	3	8.3837	cohere whisper_v2
seamless_m4t+fastconformer+hubert+cohere	4	26.569	12.4652	0.95	0.0002	0.8889	0.0001	2	1	5.7864	cohere whisper_v3
seamless_m4t+whisper_v3+hubert+cohere	4	26.5803	12.3903	0.8833	0.0015	0.7778	0.0012	4	4	8.3835	cohere whisper_v2
seamless_m4t+fastconformer+w2vbert+hubert+wav2vec_sb+cohere	6	26.5809	12.7858	0.8667	0.0023	0.7778	0.0012	4	3	8.3837	cohere whisper_v2

pool	k	ensemble_wer_vs_true	ensemble_cer_vs_t rue	ranking_spear man	ranking_exact _p	ranking_kendall _b	ranking_kendall_exact _p	n_pairs_wr ong	n_pairs_wrong_resolva ble	worst_wrong_gap _wer	worst_wrong_pair
seamless_m4t+fastconformer+w2vbert+hubert+coh ere	5	26.5904	12.5877	0.95	0.0002	0.8889	0.0001	2	1	5.7864	cohere whisper_v3
seamless_m4t+whisper_v3+cohere	3	26.5922	12.5622								
seamless_m4t+whisper_v3+w2vbert+cohere	4	26.6137	12.6761	0.9167	0.0007	0.8333	0.0004	3	3	8.3835	cohere whisper_v2
seamless_m4t+fastconformer+w2vbert+wav2vec _sb+cohere	5	26.6239	12.6343	0.8667	0.0023	0.7778	0.0012	4	3	8.3883	cohere whisper_v2
seamless_m4t+fastconformer+w2vbert+cohere	4	26.6382	12.6081	0.8667	0.0023	0.7778	0.0012	4	3	8.3837	cohere whisper_v2
seamless_m4t+fastconformer+wav2vec_sb+cohe re	4	26.6453	12.5824	0.8667	0.0023	0.7778	0.0012	4	3	8.3883	cohere whisper_v2
seamless_m4t+fastconformer+cohere	3	26.6543	12.6151								
seamless_m4t+whisper_v3+w2vbert+hubert+coh ere	5	26.6931	12.6724	0.9167	0.0007	0.8333	0.0004	3	3	8.3837	cohere whisper_v2
seamless_m4t+whisper_v3+w2vbert+hubert+wav 2vec_sb+cohere	6	26.7026	12.7477	0.8833	0.0015	0.7778	0.0012	4	4	8.3837	cohere whisper_v2
whisper_v3+fastconformer+w2vbert+cohere	4	26.8697	12.5624	0.6833	0.0252	0.5	0.0376	9	8	8.4266	cohere seamless_m4t
whisper_v3+fastconformer+cohere	3	26.8858	12.6194								
whisper_v3+fastconformer+wav2vec_sb+cohere	4	26.8866	12.6185	0.7	0.0216	0.5556	0.0223	8	8	8.4274	cohere seamless_m4t
whisper_v3+fastconformer+w2vbert+wav2vec_sb +cohere	5	26.9439	12.6638	0.7833	0.0086	0.6667	0.0063	6	6	8.4274	cohere seamless_m4t
whisper_v3+fastconformer+hubert+wav2vec_sb+ cohere	5	26.9845	12.6728	0.8333	0.0041	0.6667	0.0063	6	5	8.3821	cohere whisper_v2
whisper_v3+fastconformer+w2vbert+hubert+cohe re	5	27.0153	12.6496	0.8333	0.0041	0.6667	0.0063	6	5	8.3837	cohere whisper_v2
whisper_v3+fastconformer+hubert+cohere	4	27.0273	12.6553	0.8333	0.0041	0.6667	0.0063	6	5	8.3837	cohere whisper_v2
whisper_v3+fastconformer+w2vbert+hubert+wav2 vec_sb+cohere	6	27.1204	12.8916	0.8333	0.0041	0.6667	0.0063	6	5	8.3837	cohere whisper_v2
seamless_m4t+hubert+cohere	3	28.4852	13.8146								
seamless_m4t+whisper_v3+fastconformer	3	28.6737	13.4102								
seamless_m4t+whisper_v3+fastconformer+w2vb ert	4	28.6911	13.4075	0.9333	0.0004	0.8333	0.0004	3	2	2.9393	fastconformer whisper_v2
seamless_m4t+whisper_v3+fastconformer+huber t+wav2vec_sb	5	28.701	13.4489	0.9333	0.0004	0.8333	0.0004	3	2	2.9347	fastconformer whisper_v2
seamless_m4t+whisper_v3+fastconformer+wav2 vec_sb	4	28.713	13.4141	0.9333	0.0004	0.8333	0.0004	3	2	2.9331	fastconformer whisper_v2
seamless_m4t+whisper_v3+fastconformer+w2vb ert+wav2vec_sb	5	28.713	13.4341	0.9333	0.0004	0.8333	0.0004	3	2	2.9347	fastconformer whisper_v2
seamless_m4t+w2vbert+hubert+cohere	4	28.7413	13.9863	0.8833	0.0015	0.7778	0.0012	4	3	8.3837	cohere whisper_v2
seamless_m4t+whisper_v3+fastconformer+huber t	4	28.7866	13.3737	0.9333	0.0004	0.8333	0.0004	3	2	2.9393	fastconformer whisper_v2
seamless_m4t+whisper_v3+fastconformer+w2vb ert+hubert	5	28.7938	13.4861	0.9333	0.0004	0.8333	0.0004	3	2	2.941	fastconformer whisper_v2
seamless_m4t+hubert+wav2vec_sb+cohere	4	28.8033	14.0958	0.8833	0.0015	0.7778	0.0012	4	3	8.3837	cohere whisper_v2
seamless_m4t+whisper_v3+fastconformer+w2vb ert+hubert+wav2vec_sb	6	28.832	13.729	0.9333	0.0004	0.8333	0.0004	3	2	2.941	fastconformer whisper_v2
fastconformer+hubert+cohere	3	29.0206	13.9329								
fastconformer+hubert+wav2vec_sb+cohere	4	29.052	14.0501	0.7333	0.0156	0.6111	0.0124	7	6	8.4274	cohere seamless_m4t
whisper_v3+hubert+cohere	3	29.1464	13.656								
fastconformer+w2vbert+hubert+cohere	4	29.19	14.0611	0.7333	0.0156	0.6111	0.0124	7	6	8.4266	cohere seamless_m4t

pool	k	ensemble_wer_vs_true	ensemble_cer_vs_t_rue	ranking_spearman	ranking_exact_p	ranking_kendall_b	ranking_kendall_exact_p	n_pairs_wrong	n_pairs_wrong_resolvable	worst_wrong_gap_wer	worst_wrong_pair
seamless_m4t+w2vbert+hubert+wav2vec_sb+cohere	5	29.2593	14.4831	0.8833	0.0015	0.7778	0.0012	4	3	8.3837	cohere whisper_v2
whisper_v3+hubert+wav2vec_sb+cohere	4	29.2736	13.7986	0.75	0.0127	0.6111	0.0124	7	7	8.4266	cohere seamless_m4t
whisper_v3+w2vbert+hubert+cohere	4	29.3189	13.942	0.75	0.0127	0.6111	0.0124	7	7	8.4266	cohere seamless_m4t
fastconformer+w2vbert+hubert+wav2vec_sb+cohere	5	29.5577	14.4927	0.7333	0.0156	0.6111	0.0124	7	6	8.4266	cohere seamless_m4t
fastconformer+w2vbert+wav2vec_sb+cohere	4	29.5629	14.4431	0.5833	0.054	0.5	0.0376	9	8	27.1445	cohere hubert
whisper_v3+w2vbert+wav2vec_sb+cohere	4	29.6126	14.2692	0.6167	0.0429	0.5	0.0376	9	9	27.1443	cohere hubert
whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	5	29.6555	14.1554	0.75	0.0127	0.6111	0.0124	7	7	8.4266	cohere seamless_m4t
whisper_v3+w2vbert+cohere	3	29.9771	14.3624								
seamless_m4t+w2vbert+wav2vec_sb+cohere	4	29.9802	15.0524	0.7833	0.0086	0.6667	0.0063	6	5	27.1443	cohere hubert
fastconformer+w2vbert+cohere	3	29.9921	14.6561								
whisper_v3+wav2vec_sb+cohere	3	30.3566	15.086								
fastconformer+wav2vec_sb+cohere	3	30.4916	15.384								
seamless_m4t+w2vbert+cohere	3	30.6001	15.5509								
seamless_m4t+wav2vec_sb+cohere	3	31.142	16.4232								
seamless_m4t+fastconformer+hubert	3	31.5557	15.1582								
seamless_m4t+whisper_v3+hubert	3	31.6671	14.8964								
whisper_v3+fastconformer+hubert	3	31.6774	14.8085								
seamless_m4t+fastconformer+w2vbert+hubert	4	31.7061	15.2906	0.95	0.0002	0.8889	0.0001	2	1	2.941	fastconformer whisper_v2
seamless_m4t+fastconformer+hubert+wav2vec_sb	4	31.7688	15.3383	0.95	0.0002	0.8889	0.0001	2	1	2.9347	fastconformer whisper_v2
seamless_m4t+whisper_v3+w2vbert+hubert	4	31.8398	15.0595	0.95	0.0002	0.8889	0.0001	2	2	2.941	fastconformer whisper_v2
whisper_v3+fastconformer+hubert+wav2vec_sb	4	31.8428	14.9781	0.85	0.003	0.7222	0.0029	5	4	2.9801	fastconformer seamless_m4t
whisper_v3+fastconformer+w2vbert+hubert	4	31.8804	15.0267	0.8667	0.0023	0.7778	0.0012	4	4	2.9839	fastconformer seamless_m4t
seamless_m4t+whisper_v3+hubert+wav2vec_sb	4	31.9376	15.1229	0.95	0.0002	0.8889	0.0001	2	2	2.941	fastconformer whisper_v2
seamless_m4t+fastconformer+w2vbert+hubert+wav2vec_sb	5	32.0785	15.7065	0.95	0.0002	0.8889	0.0001	2	1	2.941	fastconformer whisper_v2
seamless_m4t+whisper_v3+w2vbert+hubert+wav2vec_sb	5	32.2098	15.4859	0.95	0.0002	0.8889	0.0001	2	2	2.941	fastconformer whisper_v2
whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb	5	32.2265	15.4578	0.8667	0.0023	0.7778	0.0012	4	4	2.9839	fastconformer seamless_m4t
whisper_v3+fastconformer+w2vbert+wav2vec_sb	4	32.4588	15.4021	0.75	0.0127	0.6667	0.0063	6	6	21.6971	fastconformer hubert
whisper_v3+fastconformer+w2vbert	3	32.5201	15.4621								
seamless_m4t+fastconformer+w2vbert+wav2vec_sb	4	32.6354	16.0558	0.8667	0.0023	0.7778	0.0012	4	3	21.6971	fastconformer hubert
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	32.7875	15.8641	0.8833	0.0015	0.7778	0.0012	4	4	21.3578	whisper_v3 hubert
whisper_v3+fastconformer+wav2vec_sb	3	33.1153	16.1872								
seamless_m4t+whisper_v3+w2vbert	3	33.2546	16.3084								
seamless_m4t+fastconformer+w2vbert	3	33.4105	16.7397								
seamless_m4t+fastconformer+wav2vec_sb	3	34.1514	17.6679								
seamless_m4t+whisper_v3+wav2vec_sb	3	34.1784	17.323								

pool	k	ensemble_wer_vs_true	ensemble_cer_vs_true	ranking_spearman	ranking_exact_p	ranking_kendall_b	ranking_kendall_exact_p	n_pairs_wrong	n_pairs_wrong_resolvable	worst_wrong_gap_wer	worst_wrong_pair
w2vbert+hubert+cohere	3	41.2595	21.2686								
w2vbert+hubert+wav2vec_sb+cohere	4	41.5507	21.3677	0.7333	0.0156	0.6667	0.0063	6	5	27.1443	cohere hubert
fastconformer+w2vbert+hubert	3	43.1907	21.713								
whisper_v3+w2vbert+hubert	3	43.4055	21.6149								
hubert+wav2vec_sb+cohere	3	43.4413	23.4775								
fastconformer+w2vbert+hubert+wav2vec_sb	4	43.6991	22.0427	0.8	0.0069	0.7222	0.0029	5	4	21.6971	fastconformer hubert
whisper_v3+w2vbert+hubert+wav2vec_sb	4	43.7994	21.9465	0.8	0.0069	0.6667	0.0063	6	6	21.3578	whisper_v3 hubert
seamless_m4t+w2vbert+hubert	3	44.3293	22.8358								
seamless_m4t+w2vbert+hubert+wav2vec_sb	4	44.4033	22.4705	0.8833	0.0015	0.7222	0.0029	5	4	34.11	seamless_m4t w2vbert
fastconformer+hubert+wav2vec_sb	3	45.2432	23.8145								
whisper_v3+hubert+wav2vec_sb	3	45.5539	23.6824								
seamless_m4t+hubert+wav2vec_sb	3	46.6234	25.1048								
w2vbert+wav2vec_sb+cohere	3	52.6581	29.5503								
whisper_v3+w2vbert+wav2vec_sb	3	54.0856	29.6408								
fastconformer+w2vbert+wav2vec_sb	3	54.1743	29.6942								
seamless_m4t+w2vbert+wav2vec_sb	3	55.3914	31.0528								
w2vbert+hubert+wav2vec_sb	3	61.4738	32.7201								

step6_wrong_pairs_validated_ensemble_rdi

pool	k	pair	true_winner	pred_winner	wer_true_a	wer_true_b	wer_pseudo_a	wer_pseudo_b	gap_wer_true	gap_wer_pseudo	human_resolvable
seamless_m4t+w2vbert+hubert+wav2vec_sb	4	seamless_m4t w2vbert	seamless_m4t	w2vbert	34.912	69.022	68.9853	67.7593	34.11	1.2261	True
fastconformer+w2vbert+wav2vec_sb+cohere	4	cohere hubert	cohere	hubert	26.4808	53.6252	56.2578	49.5732	27.1445	6.6846	True
seamless_m4t+w2vbert+wav2vec_sb+cohere	4	cohere hubert	cohere	hubert	26.4854	53.6297	57.3154	50.7708	27.1443	6.5446	True
whisper_v3+w2vbert+wav2vec_sb+cohere	4	cohere hubert	cohere	hubert	26.4854	53.6297	55.7084	49.4458	27.1443	6.2627	True
w2vbert+hubert+wav2vec_sb+cohere	4	cohere hubert	cohere	hubert	26.4854	53.6297	67.0705	62.8443	27.1443	4.2262	True
seamless_m4t+fastconformer+w2vbert+wav2vec_sb	4	fastconformer hubert	fastconformer	hubert	31.9281	53.6252	56.8546	50.0998	21.6971	6.7548	True
whisper_v3+fastconformer+w2vbert+wav2vec_sb	4	fastconformer hubert	fastconformer	hubert	31.9281	53.6252	54.8739	49.1006	21.6971	5.7733	True
fastconformer+w2vbert+hubert+wav2vec_sb	4	fastconformer hubert	fastconformer	hubert	31.9281	53.6252	64.7358	62.5063	21.6971	2.2295	True
fastconformer+w2vbert+wav2vec_sb+cohere	4	fastconformer hubert	fastconformer	hubert	31.9281	53.6252	54.5584	49.5732	21.6971	4.9852	True
whisper_v3+fastconformer+w2vbert+wav2vec_sb	4	whisper_v3 hubert	whisper_v3	hubert	32.2654	53.6252	55.378	49.1006	21.3598	6.2774	True
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	whisper_v3 hubert	whisper_v3	hubert	32.2718	53.6297	56.7804	50.1672	21.3578	6.6133	True
whisper_v3+w2vbert+hubert+wav2vec_sb	4	whisper_v3 hubert	whisper_v3	hubert	32.2718	53.6297	64.8989	62.2052	21.3578	2.6937	True
whisper_v3+w2vbert+wav2vec_sb+cohere	4	whisper_v3 hubert	whisper_v3	hubert	32.2718	53.6297	54.7114	49.4458	21.3578	5.2656	True
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	seamless_m4t hubert	seamless_m4t	hubert	34.912	53.6297	58.6188	50.1672	18.7176	8.4516	True
seamless_m4t+w2vbert+hubert+wav2vec_sb	4	seamless_m4t hubert	seamless_m4t	hubert	34.912	53.6297	68.9853	63.9038	18.7176	5.0815	True
seamless_m4t+w2vbert+wav2vec_sb+cohere	4	seamless_m4t hubert	seamless_m4t	hubert	34.912	53.6297	57.9983	50.7708	18.7176	7.2275	True
seamless_m4t+fastconformer+w2vbert+wav2vec_sb	4	seamless_m4t hubert	seamless_m4t	hubert	34.9082	53.6252	59.231	50.0998	18.717	9.1312	True
whisper_v3+fastconformer+wav2vec_sb+cohere	4	cohere seamless_m4t	cohere	seamless_m4t	26.4808	34.9082	25.9678	24.8442	8.4274	1.1235	True
fastconformer+w2vbert+wav2vec_sb+cohere	4	cohere seamless_m4t	cohere	seamless_m4t	26.4808	34.9082	56.2578	28.369	8.4274	27.8888	True
fastconformer+hubert+wav2vec_sb+cohere	4	cohere seamless_m4t	cohere	seamless_m4t	26.4808	34.9082	43.5251	28.0781	8.4274	15.447	True
whisper_v3+fastconformer+w2vbert+wav2vec_sb+cohere	5	cohere seamless_m4t	cohere	seamless_m4t	26.4808	34.9082	25.066	24.9752	8.4274	0.0907	True

pool	k	pair	true_winner	pred_winner	wer_true_a	wer_true_b	wer_pseudo_a	wer_pseudo_b	gap_wer_true	gap_wer_pseudo	human_resolvable
whisper_v3+fastconformer+w2vbert+cohere	4	cohere seamless_m4t	cohere	seamless_m4t	26.4854	34.912	25.2382	24.877	8.4266	0.3612	True
whisper_v3+w2vbert+hubert+cohere	4	cohere seamless_m4t	cohere	seamless_m4t	26.4854	34.912	40.7732	27.6905	8.4266	13.0828	True
whisper_v3+w2vbert+wav2vec_sb+cohere	4	cohere seamless_m4t	cohere	seamless_m4t	26.4854	34.912	55.7084	27.9761	8.4266	27.7323	True
whisper_v3+hubert+wav2vec_sb+cohere	4	cohere seamless_m4t	cohere	seamless_m4t	26.4854	34.912	43.4617	27.7805	8.4266	15.6812	True
fastconformer+w2vbert+hubert+cohere	4	cohere seamless_m4t	cohere	seamless_m4t	26.4854	34.912	40.9936	28.1145	8.4266	12.8791	True
w2vbert+hubert+wav2vec_sb+cohere	4	cohere seamless_m4t	cohere	seamless_m4t	26.4854	34.912	67.0705	43.8611	8.4266	23.2094	True
whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	5	cohere seamless_m4t	cohere	seamless_m4t	26.4854	34.912	41.1885	28.4343	8.4266	12.7543	True
fastconformer+w2vbert+hubert+wav2vec_sb+cohere	5	cohere seamless_m4t	cohere	seamless_m4t	26.4854	34.912	41.3407	28.9371	8.4266	12.4036	True
seamless_m4t+fastconformer+wav2vec_sb+cohere	4	cohere whisper_v2	cohere	whisper_v2	26.4808	34.8691	26.8928	24.785	8.3883	2.1078	True
seamless_m4t+fastconformer+w2vbert+wav2vec_sb+cohere	5	cohere whisper_v2	cohere	whisper_v2	26.4808	34.8691	24.9706	24.7647	8.3883	0.2059	True
seamless_m4t+fastconformer+w2vbert+cohere	4	cohere whisper_v2	cohere	whisper_v2	26.4854	34.8691	25.9528	24.7778	8.3837	1.1749	True
seamless_m4t+w2vbert+hubert+cohere	4	cohere whisper_v2	cohere	whisper_v2	26.4854	34.8691	41.8456	27.0721	8.3837	14.7735	True
seamless_m4t+w2vbert+wav2vec_sb+cohere	4	cohere whisper_v2	cohere	whisper_v2	26.4854	34.8691	57.3154	28.2442	8.3837	29.0712	True
seamless_m4t+hubert+wav2vec_sb+cohere	4	cohere whisper_v2	cohere	whisper_v2	26.4854	34.8691	44.537	27.1749	8.3837	17.362	True
whisper_v3+fastconformer+w2vbert+cohere	4	cohere whisper_v2	cohere	whisper_v2	26.4854	34.8691	25.2382	23.1398	8.3837	2.0983	True
whisper_v3+fastconformer+hubert+cohere	4	cohere whisper_v2	cohere	whisper_v2	26.4854	34.8691	24.3416	23.3054	8.3837	1.0363	True
whisper_v3+w2vbert+hubert+wav2vec_sb	4	cohere whisper_v2	cohere	whisper_v2	26.4854	34.8691	41.1885	39.7698	8.3837	1.4187	True
whisper_v3+w2vbert+hubert+cohere	4	cohere whisper_v2	cohere	whisper_v2	26.4854	34.8691	40.7732	23.974	8.3837	16.7992	True
whisper_v3+w2vbert+wav2vec_sb+cohere	4	cohere whisper_v2	cohere	whisper_v2	26.4854	34.8691	55.7084	24.254	8.3837	31.4544	True
whisper_v3+hubert+wav2vec_sb+cohere	4	cohere whisper_v2	cohere	whisper_v2	26.4854	34.8691	43.4617	23.9378	8.3837	19.5238	True
fastconformer+w2vbert+hubert+cohere	4	cohere whisper_v2	cohere	whisper_v2	26.4854	34.8691	40.9936	27.3268	8.3837	13.6668	True
w2vbert+hubert+wav2vec_sb+cohere	4	cohere whisper_v2	cohere	whisper_v2	26.4854	34.8691	67.0705	41.8257	8.3837	25.2448	True
seamless_m4t+whisper_v3+w2vbert+hubert+cohere	5	cohere whisper_v2	cohere	whisper_v2	26.4854	34.8691	24.1524	22.7898	8.3837	1.3626	True
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb+cohere	5	cohere whisper_v2	cohere	whisper_v2	26.4854	34.8691	25.0788	22.9072	8.3837	2.1716	True
seamless_m4t+whisper_v3+hubert+wav2vec_sb+cohere	5	cohere whisper_v2	cohere	whisper_v2	26.4854	34.8691	24.2065	22.7724	8.3837	1.4341	True
seamless_m4t+w2vbert+hubert+wav2vec_sb+cohere	5	cohere whisper_v2	cohere	whisper_v2	26.4854	34.8691	41.7585	27.626	8.3837	14.1325	True
whisper_v3+fastconformer+w2vbert+hubert+cohere	5	cohere whisper_v2	cohere	whisper_v2	26.4854	34.8691	24.5666	23.1072	8.3837	1.4594	True
whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	5	cohere whisper_v2	cohere	whisper_v2	26.4854	34.8691	41.1885	24.5139	8.3837	16.6746	True
fastconformer+w2vbert+hubert+wav2vec_sb+cohere	5	cohere whisper_v2	cohere	whisper_v2	26.4854	34.8691	41.3407	27.9171	8.3837	13.4236	True
seamless_m4t+whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	6	cohere whisper_v2	cohere	whisper_v2	26.4854	34.8691	25.0133	22.9822	8.3837	2.0311	True
seamless_m4t+fastconformer+w2vbert+hubert+wav2vec_sb+cohere	6	cohere whisper_v2	cohere	whisper_v2	26.4854	34.8691	25.1626	24.8616	8.3837	0.3009	True
whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb+cohere	6	cohere whisper_v2	cohere	whisper_v2	26.4854	34.8691	25.4146	23.2406	8.3837	2.1739	True
seamless_m4t+whisper_v3+w2vbert+cohere	4	cohere whisper_v2	cohere	whisper_v2	26.4872	34.8706	25.7686	22.9131	8.3835	2.8556	True
seamless_m4t+whisper_v3+hubert+cohere	4	cohere whisper_v2	cohere	whisper_v2	26.4872	34.8706	23.9177	23.0408	8.3835	0.8769	True
seamless_m4t+whisper_v3+wav2vec_sb+cohere	4	cohere whisper_v2	cohere	whisper_v2	26.4872	34.8706	26.672	22.9337	8.3835	3.7383	True
whisper_v3+fastconformer+wav2vec_sb+cohere	4	cohere whisper_v2	cohere	whisper_v2	26.4808	34.8628	25.9678	23.2018	8.3821	2.766	True
fastconformer+w2vbert+wav2vec_sb+cohere	4	cohere whisper_v2	cohere	whisper_v2	26.4808	34.8628	56.2578	27.9062	8.3821	28.3516	True
fastconformer+hubert+wav2vec_sb+cohere	4	cohere whisper_v2	cohere	whisper_v2	26.4808	34.8628	43.5251	27.3314	8.3821	16.1937	True
whisper_v3+fastconformer+w2vbert+wav2vec_sb+cohere	5	cohere whisper_v2	cohere	whisper_v2	26.4808	34.8628	25.066	23.149	8.3821	1.9169	True
whisper_v3+fastconformer+hubert+wav2vec_sb+cohere	5	cohere whisper_v2	cohere	whisper_v2	26.4808	34.8628	24.4788	23.0699	8.3821	1.4089	True
seamless_m4t+fastconformer+wav2vec_sb+cohere	4	cohere whisper_v3	cohere	whisper_v3	26.4808	32.2718	26.8928	21.3904	5.7911	5.5024	True
seamless_m4t+fastconformer+w2vbert+wav2vec_sb+cohere	5	cohere whisper_v3	cohere	whisper_v3	26.4808	32.2718	24.9706	21.4595	5.7911	3.511	True

pool	k	pair	true_winner	pred_winner	wer_true_a	wer_true_b	wer_pseudo_a	wer_pseudo_b	gap_wer_true	gap_wer_pseudo	human_resolvable
seamless_m4t+fastconformer+hubert+wav2vec_sb+cohere	5	cohere whisper_v3	cohere	whisper_v3	26.4808	32.2718	24.2474	21.4867	5.7911	2.7607	True
seamless_m4t+fastconformer+w2vbert+cohere	4	cohere whisper_v3	cohere	whisper_v3	26.4854	32.2718	25.9528	21.4151	5.7864	4.5377	True
seamless_m4t+fastconformer+hubert+cohere	4	cohere whisper_v3	cohere	whisper_v3	26.4854	32.2718	24.0104	21.4903	5.7864	2.5202	True
seamless_m4t+w2vbert+hubert+wav2vec_sb	4	cohere whisper_v3	cohere	whisper_v3	26.4854	32.2718	41.7585	41.1923	5.7864	0.5662	True
seamless_m4t+w2vbert+hubert+cohere	4	cohere whisper_v3	cohere	whisper_v3	26.4854	32.2718	41.8456	24.2306	5.7864	17.615	True
seamless_m4t+w2vbert+wav2vec_sb+cohere	4	cohere whisper_v3	cohere	whisper_v3	26.4854	32.2718	57.3154	25.4952	5.7864	31.8202	True
seamless_m4t+hubert+wav2vec_sb+cohere	4	cohere whisper_v3	cohere	whisper_v3	26.4854	32.2718	44.537	24.2918	5.7864	20.2452	True
whisper_v3+fastconformer+w2vbert+cohere	4	cohere whisper_v3	cohere	whisper_v3	26.4854	32.2718	25.2382	25.1236	5.7864	0.1146	True
whisper_v3+fastconformer+hubert+cohere	4	cohere whisper_v3	cohere	whisper_v3	26.4854	32.2718	24.3416	23.9625	5.7864	0.3791	True
whisper_v3+w2vbert+hubert+cohere	4	cohere whisper_v3	cohere	whisper_v3	26.4854	32.2718	40.7732	39.0443	5.7864	1.729	True
whisper_v3+w2vbert+wav2vec_sb+cohere	4	cohere whisper_v3	cohere	whisper_v3	26.4854	32.2718	55.7084	54.7114	5.7864	0.997	True
whisper_v3+hubert+wav2vec_sb+cohere	4	cohere whisper_v3	cohere	whisper_v3	26.4854	32.2718	43.4617	41.7521	5.7864	1.7096	True
fastconformer+w2vbert+hubert+wav2vec_sb	4	cohere whisper_v3	cohere	whisper_v3	26.4854	32.2718	41.3407	40.5021	5.7864	0.8386	True
fastconformer+w2vbert+hubert+cohere	4	cohere whisper_v3	cohere	whisper_v3	26.4854	32.2718	40.9936	24.157	5.7864	16.8366	True
w2vbert+hubert+wav2vec_sb+cohere	4	cohere whisper_v3	cohere	whisper_v3	26.4854	32.2718	67.0705	39.523	5.7864	27.5475	True
seamless_m4t+fastconformer+w2vbert+hubert+cohere	5	cohere whisper_v3	cohere	whisper_v3	26.4854	32.2718	24.1961	21.4685	5.7864	2.7277	True
seamless_m4t+w2vbert+hubert+wav2vec_sb+cohere	5	cohere whisper_v3	cohere	whisper_v3	26.4854	32.2718	41.7585	24.6427	5.7864	17.1158	True
whisper_v3+fastconformer+w2vbert+hubert+cohere	5	cohere whisper_v3	cohere	whisper_v3	26.4854	32.2718	24.5666	24.157	5.7864	0.4096	True
whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	5	cohere whisper_v3	cohere	whisper_v3	26.4854	32.2718	41.1885	39.523	5.7864	1.6656	True
fastconformer+w2vbert+hubert+wav2vec_sb+cohere	5	cohere whisper_v3	cohere	whisper_v3	26.4854	32.2718	41.3407	24.7311	5.7864	16.6096	True
seamless_m4t+whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	6	cohere whisper_v3	cohere	whisper_v3	26.4854	32.2718	25.0133	24.6427	5.7864	0.3706	True
seamless_m4t+fastconformer+w2vbert+hubert+wav2vec_sb+cohere	6	cohere whisper_v3	cohere	whisper_v3	26.4854	32.2718	25.1626	21.574	5.7864	3.5886	True
whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb+cohere	6	cohere whisper_v3	cohere	whisper_v3	26.4854	32.2718	25.4146	24.7311	5.7864	0.6835	True
seamless_m4t+whisper_v3+hubert+cohere	4	cohere whisper_v3	cohere	whisper_v3	26.4872	32.2735	23.9177	23.885	5.7863	0.0327	True
whisper_v3+fastconformer+wav2vec_sb+cohere	4	cohere whisper_v3	cohere	whisper_v3	26.4808	32.2654	25.9678	25.8564	5.7846	0.1113	True
fastconformer+w2vbert+wav2vec_sb+cohere	4	cohere whisper_v3	cohere	whisper_v3	26.4808	32.2654	56.2578	24.8202	5.7846	31.4376	True
fastconformer+hubert+wav2vec_sb+cohere	4	cohere whisper_v3	cohere	whisper_v3	26.4808	32.2654	43.5251	24.0835	5.7846	19.4416	True
whisper_v3+fastconformer+w2vbert+wav2vec_sb+cohere	5	cohere whisper_v3	cohere	whisper_v3	26.4808	32.2654	25.066	24.8202	5.7846	0.2458	True
whisper_v3+fastconformer+hubert+wav2vec_sb+cohere	5	cohere whisper_v3	cohere	whisper_v3	26.4808	32.2654	24.4788	24.0835	5.7846	0.3952	True
whisper_v3+fastconformer+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	26.4808	31.9297	25.9678	25.6617	5.449	0.306	True
fastconformer+w2vbert+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	26.4808	31.9281	56.2578	54.5584	5.4473	1.6994	True
fastconformer+hubert+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	26.4808	31.9281	43.5251	41.6688	5.4473	1.8563	True
whisper_v3+fastconformer+w2vbert+wav2vec_sb+cohere	5	cohere fastconformer	cohere	fastconformer	26.4808	31.9281	25.066	24.8022	5.4473	0.2637	True
whisper_v3+fastconformer+hubert+wav2vec_sb+cohere	5	cohere fastconformer	cohere	fastconformer	26.4808	31.9281	24.4788	24.2109	5.4473	0.2678	True
whisper_v3+fastconformer+w2vbert+cohere	4	cohere fastconformer	cohere	fastconformer	26.4854	31.9297	25.2382	25.1507	5.4443	0.0875	True
whisper_v3+fastconformer+hubert+cohere	4	cohere fastconformer	cohere	fastconformer	26.4854	31.9297	24.3416	24.0121	5.4443	0.3295	True
seamless_m4t+w2vbert+hubert+wav2vec_sb	4	cohere fastconformer	cohere	fastconformer	26.4854	31.9281	41.7585	41.4916	5.4427	0.2669	True
seamless_m4t+w2vbert+hubert+cohere	4	cohere fastconformer	cohere	fastconformer	26.4854	31.9281	41.8456	24.567	5.4427	17.2786	True
seamless_m4t+w2vbert+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	26.4854	31.9281	57.3154	25.76	5.4427	31.5553	True
seamless_m4t+hubert+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	26.4854	31.9281	44.537	24.6145	5.4427	19.9225	True
whisper_v3+w2vbert+hubert+wav2vec_sb	4	cohere fastconformer	cohere	fastconformer	26.4854	31.9281	41.1885	40.3878	5.4427	0.8007	True
whisper_v3+w2vbert+hubert+cohere	4	cohere fastconformer	cohere	fastconformer	26.4854	31.9281	40.7732	24.3423	5.4427	16.4309	True

pool	k	pair	true_winner	pred_winner	wer_true_a	wer_true_b	wer_pseudo_a	wer_pseudo_b	gap_wer_true	gap_wer_pseudo	human_resolvable
whisper_v3+w2vbert+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	26.4854	31.9281	55.7084	24.8022	5.4427	30.9062	True
whisper_v3+hubert+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	26.4854	31.9281	43.4617	24.2109	5.4427	19.2507	True
fastconformer+w2vbert+hubert+cohere	4	cohere fastconformer	cohere	fastconformer	26.4854	31.9281	40.9936	39.0643	5.4427	1.9293	True
w2vbert+hubert+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	26.4854	31.9281	67.0705	39.5679	5.4427	27.5026	True
seamless_m4t+whisper_v3+w2vbert+hubert+cohere	5	cohere fastconformer	cohere	fastconformer	26.4854	31.9281	24.1524	21.8521	5.4427	2.3004	True
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb+cohere	5	cohere fastconformer	cohere	fastconformer	26.4854	31.9281	25.0788	21.8689	5.4427	3.2098	True
seamless_m4t+whisper_v3+hubert+wav2vec_sb+cohere	5	cohere fastconformer	cohere	fastconformer	26.4854	31.9281	24.2065	21.8224	5.4427	2.3841	True
seamless_m4t+w2vbert+hubert+wav2vec_sb+cohere	5	cohere fastconformer	cohere	fastconformer	26.4854	31.9281	41.7585	25.1784	5.4427	16.5802	True
whisper_v3+fastconformer+w2vbert+hubert+cohere	5	cohere fastconformer	cohere	fastconformer	26.4854	31.9281	24.5666	24.3423	5.4427	0.2243	True
whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	5	cohere fastconformer	cohere	fastconformer	26.4854	31.9281	41.1885	24.817	5.4427	16.3716	True
fastconformer+w2vbert+hubert+wav2vec_sb+cohere	5	cohere fastconformer	cohere	fastconformer	26.4854	31.9281	41.3407	39.5679	5.4427	1.7728	True
seamless_m4t+whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	6	cohere fastconformer	cohere	fastconformer	26.4854	31.9281	25.0133	21.9683	5.4427	3.0451	True
whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb+cohere	6	cohere fastconformer	cohere	fastconformer	26.4854	31.9281	25.4146	24.817	5.4427	0.5976	True
seamless_m4t+whisper_v3+w2vbert+cohere	4	cohere fastconformer	cohere	fastconformer	26.4872	31.9297	25.7686	21.7976	5.4426	3.971	True
seamless_m4t+whisper_v3+hubert+cohere	4	cohere fastconformer	cohere	fastconformer	26.4872	31.9297	23.9177	21.9134	5.4426	2.0043	True
seamless_m4t+whisper_v3+wav2vec_sb+cohere	4	cohere fastconformer	cohere	fastconformer	26.4872	31.9297	26.672	21.7557	5.4426	4.9163	True
whisper_v3+fastconformer+w2vbert+hubert	4	fastconformer seamless_m4t	fastconformer	seamless_m4t	31.9281	34.912	39.9769	29.7921	2.9839	10.1848	True
fastconformer+w2vbert+hubert+wav2vec_sb	4	fastconformer seamless_m4t	fastconformer	seamless_m4t	31.9281	34.912	64.7358	45.3226	2.9839	19.4132	True
fastconformer+w2vbert+hubert+cohere	4	fastconformer seamless_m4t	fastconformer	seamless_m4t	31.9281	34.912	39.0643	28.1145	2.9839	10.9498	True
whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb	5	fastconformer seamless_m4t	fastconformer	seamless_m4t	31.9281	34.912	40.3878	30.519	2.9839	9.8688	True
fastconformer+w2vbert+hubert+wav2vec_sb+cohere	5	fastconformer seamless_m4t	fastconformer	seamless_m4t	31.9281	34.912	39.5679	28.9371	2.9839	10.6309	True
whisper_v3+fastconformer+w2vbert+cohere	4	fastconformer seamless_m4t	fastconformer	seamless_m4t	31.9297	34.912	25.1507	24.877	2.9823	0.2737	True
whisper_v3+fastconformer+w2vbert+wav2vec_sb	4	fastconformer seamless_m4t	fastconformer	seamless_m4t	31.9281	34.9082	54.8739	30.1959	2.9801	24.678	True
whisper_v3+fastconformer+hubert+wav2vec_sb	4	fastconformer seamless_m4t	fastconformer	seamless_m4t	31.9281	34.9082	42.5769	29.7106	2.9801	12.8663	True
fastconformer+w2vbert+wav2vec_sb+cohere	4	fastconformer seamless_m4t	fastconformer	seamless_m4t	31.9281	34.9082	54.5584	28.369	2.9801	26.1894	True
fastconformer+hubert+wav2vec_sb+cohere	4	fastconformer seamless_m4t	fastconformer	seamless_m4t	31.9281	34.9082	41.6688	28.0781	2.9801	13.5907	True
whisper_v3+fastconformer+wav2vec_sb+cohere	4	fastconformer seamless_m4t	fastconformer	seamless_m4t	31.9297	34.9082	25.6617	24.8442	2.9785	0.8175	True
seamless_m4t+whisper_v3+w2vbert+hubert	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8691	26.0231	25.1695	2.941	0.8536	True
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8691	27.1065	26.1516	2.941	0.9549	True
seamless_m4t+whisper_v3+hubert+wav2vec_sb	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8691	25.9128	25.1344	2.941	0.7784	True
seamless_m4t+fastconformer+w2vbert+hubert	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8691	41.4865	28.3263	2.941	13.1602	True
whisper_v3+fastconformer+w2vbert+hubert	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8691	39.9769	25.5416	2.941	14.4352	True
whisper_v3+w2vbert+hubert+wav2vec_sb	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8691	40.3878	39.7698	2.941	0.618	True
whisper_v3+w2vbert+hubert+cohere	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8691	24.3423	23.974	2.941	0.3683	True
whisper_v3+w2vbert+wav2vec_sb+cohere	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8691	24.8022	24.254	2.941	0.5482	True
whisper_v3+hubert+wav2vec_sb+cohere	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8691	24.2109	23.9378	2.941	0.2731	True
fastconformer+w2vbert+hubert+wav2vec_sb	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8691	64.7358	42.699	2.941	22.0368	True
fastconformer+w2vbert+hubert+cohere	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8691	39.0643	27.3268	2.941	11.7375	True
seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert	5	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8691	26.0231	23.4766	2.941	2.5465	True
seamless_m4t+whisper_v3+w2vbert+hubert+wav2vec_sb	5	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8691	26.6374	25.8133	2.941	0.8242	True
seamless_m4t+fastconformer+w2vbert+hubert+wav2vec_sb	5	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8691	41.4916	28.9426	2.941	12.549	True
seamless_m4t+fastconformer+w2vbert+wav2vec_sb+cohere	5	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8691	25.76	24.7647	2.941	0.9953	True
whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb	5	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8691	40.3878	26.1654	2.941	14.2224	True

pool	k	pair	true_winner	pred_winner	wer_true_a	wer_true_b	wer_pseudo_a	wer_pseudo_b	gap_wer_true	gap_wer_pseudo	human_resolvable
whisper_v3+fastconformer+w2vbert+hubert+cohere	5	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8691	24.3423	23.1072	2.941	1.2351	True
whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	5	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8691	24.817	24.5139	2.941	0.3031	True
fastconformer+w2vbert+hubert+wav2vec_sb+cohere	5	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8691	39.5679	27.9171	2.941	11.6508	True
seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb	6	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8691	26.6374	23.6962	2.941	2.9412	True
seamless_m4t+fastconformer+w2vbert+hubert+wav2vec_sb+cohere	6	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8691	25.1784	24.8616	2.941	0.3167	True
whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb+cohere	6	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8691	24.817	23.2406	2.941	1.5763	True
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9297	34.8691	27.676	23.4441	2.9393	4.2319	True
seamless_m4t+whisper_v3+fastconformer+hubert	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9297	34.8691	25.592	23.5927	2.9393	1.9993	True
seamless_m4t+fastconformer+w2vbert+cohere	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9297	34.8691	26.6499	24.7778	2.9393	1.8721	True
seamless_m4t+fastconformer+wav2vec_sb+cohere	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9297	34.8691	27.2652	24.785	2.9393	2.4802	True
whisper_v3+fastconformer+w2vbert+cohere	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9297	34.8691	25.1507	23.1398	2.9393	2.0109	True
whisper_v3+fastconformer+hubert+cohere	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9297	34.8691	24.0121	23.3054	2.9393	0.7067	True
seamless_m4t+fastconformer+w2vbert+wav2vec_sb	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8628	56.8546	29.4584	2.9347	27.3962	True
seamless_m4t+fastconformer+hubert+wav2vec_sb	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8628	44.2064	28.3127	2.9347	15.8937	True
whisper_v3+fastconformer+w2vbert+wav2vec_sb	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8628	54.8739	25.7995	2.9347	29.0744	True
whisper_v3+fastconformer+hubert+wav2vec_sb	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8628	42.5769	25.2752	2.9347	17.3017	True
fastconformer+w2vbert+wav2vec_sb+cohere	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8628	54.5584	27.9062	2.9347	26.6522	True
fastconformer+hubert+wav2vec_sb+cohere	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8628	41.6688	27.3314	2.9347	14.3374	True
seamless_m4t+whisper_v3+fastconformer+w2vbert+wav2vec_sb	5	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8628	27.1065	23.4845	2.9347	3.622	True
seamless_m4t+whisper_v3+fastconformer+hubert+wav2vec_sb	5	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8628	25.9128	23.4478	2.9347	2.465	True
whisper_v3+fastconformer+w2vbert+wav2vec_sb+cohere	5	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8628	24.8022	23.149	2.9347	1.6532	True
whisper_v3+fastconformer+hubert+wav2vec_sb+cohere	5	fastconformer whisper_v2	fastconformer	whisper_v2	31.9281	34.8628	24.2109	23.0699	2.9347	1.1411	True
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9297	34.8628	28.7067	23.4223	2.9331	5.2844	True
whisper_v3+fastconformer+wav2vec_sb+cohere	4	fastconformer whisper_v2	fastconformer	whisper_v2	31.9297	34.8628	25.6617	23.2018	2.9331	2.4599	True
whisper_v3+fastconformer+w2vbert+wav2vec_sb	4	whisper_v3 seamless_m4t	whisper_v3	seamless_m4t	32.2654	34.9082	55.378	30.1959	2.6428	25.1822	True
whisper_v3+fastconformer+hubert+wav2vec_sb	4	whisper_v3 seamless_m4t	whisper_v3	seamless_m4t	32.2654	34.9082	42.4835	29.7106	2.6428	12.7729	True
whisper_v3+fastconformer+wav2vec_sb+cohere	4	whisper_v3 seamless_m4t	whisper_v3	seamless_m4t	32.2654	34.9082	25.8564	24.8442	2.6428	1.0122	True
whisper_v3+fastconformer+w2vbert+hubert	4	whisper_v3 seamless_m4t	whisper_v3	seamless_m4t	32.2718	34.912	40.132	29.7921	2.6402	10.3399	True
whisper_v3+fastconformer+w2vbert+cohere	4	whisper_v3 seamless_m4t	whisper_v3	seamless_m4t	32.2718	34.912	25.1236	24.877	2.6402	0.2466	True
whisper_v3+w2vbert+hubert+wav2vec_sb	4	whisper_v3 seamless_m4t	whisper_v3	seamless_m4t	32.2718	34.912	64.8989	45.0519	2.6402	19.847	True
whisper_v3+w2vbert+hubert+cohere	4	whisper_v3 seamless_m4t	whisper_v3	seamless_m4t	32.2718	34.912	39.0443	27.6905	2.6402	11.3538	True
whisper_v3+w2vbert+wav2vec_sb+cohere	4	whisper_v3 seamless_m4t	whisper_v3	seamless_m4t	32.2718	34.912	54.7114	27.9761	2.6402	26.7353	True
whisper_v3+hubert+wav2vec_sb+cohere	4	whisper_v3 seamless_m4t	whisper_v3	seamless_m4t	32.2718	34.912	41.7521	27.7805	2.6402	13.9716	True
whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb	5	whisper_v3 seamless_m4t	whisper_v3	seamless_m4t	32.2718	34.912	40.5021	30.519	2.6402	9.9831	True
whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	5	whisper_v3 seamless_m4t	whisper_v3	seamless_m4t	32.2718	34.912	39.523	28.4343	2.6402	11.0887	True
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2654	34.8628	28.547	23.4223	2.5975	5.1247	True
whisper_v3+fastconformer+w2vbert+wav2vec_sb	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2654	34.8628	55.378	25.7995	2.5975	29.5785	True
whisper_v3+fastconformer+hubert+wav2vec_sb	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2654	34.8628	42.4835	25.2752	2.5975	17.2083	True
whisper_v3+fastconformer+wav2vec_sb+cohere	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2654	34.8628	25.8564	23.2018	2.5975	2.6547	True
seamless_m4t+whisper_v3+fastconformer+w2vbert+wav2vec_sb	5	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2654	34.8628	26.6508	23.4845	2.5975	3.1663	True
seamless_m4t+whisper_v3+fastconformer+hubert+wav2vec_sb	5	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2654	34.8628	25.5286	23.4478	2.5975	2.0809	True
whisper_v3+fastconformer+w2vbert+wav2vec_sb+cohere	5	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2654	34.8628	24.8202	23.149	2.5975	1.6711	True

pool	k	pair	true_winner	pred_winner	wer_true_a	wer_true_b	wer_pseudo_a	wer_pseudo_b	gap_wer_true	gap_wer_pseudo	human_resolvable
whisper_v3+fastconformer+hubert+wav2vec_sb+cohere	5	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2654	34.8628	24.0835	23.0699	2.5975	1.0137	True
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	27.4913	23.4441	2.5972	4.0471	True
seamless_m4t+whisper_v3+fastconformer+hubert	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	25.1931	23.5927	2.5972	1.6004	True
seamless_m4t+whisper_v3+w2vbert+hubert	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	41.2605	25.1695	2.5972	16.091	True
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	56.7804	26.1516	2.5972	30.6289	True
seamless_m4t+whisper_v3+hubert+wav2vec_sb	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	43.9263	25.1344	2.5972	18.792	True
whisper_v3+fastconformer+w2vbert+hubert	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	40.132	25.5416	2.5972	14.5904	True
whisper_v3+fastconformer+w2vbert+cohere	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	25.1236	23.1398	2.5972	1.9837	True
whisper_v3+fastconformer+hubert+cohere	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	23.9625	23.3054	2.5972	0.6571	True
whisper_v3+w2vbert+hubert+wav2vec_sb	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	64.8989	39.7698	2.5972	25.1291	True
whisper_v3+w2vbert+hubert+cohere	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	39.0443	23.974	2.5972	15.0702	True
whisper_v3+w2vbert+wav2vec_sb+cohere	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	54.7114	24.254	2.5972	30.4574	True
whisper_v3+hubert+wav2vec_sb+cohere	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	41.7521	23.9378	2.5972	17.8142	True
seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert	5	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	25.5054	23.4766	2.5972	2.0289	True
seamless_m4t+whisper_v3+w2vbert+hubert+wav2vec_sb	5	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	41.1923	25.8133	2.5972	15.3791	True
seamless_m4t+whisper_v3+w2vbert+hubert+cohere	5	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	24.2306	22.7898	2.5972	1.4407	True
seamless_m4t+whisper_v3+w2vbert+wav2vec_sb+cohere	5	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	25.4952	22.9072	2.5972	2.5879	True
seamless_m4t+whisper_v3+hubert+wav2vec_sb+cohere	5	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	24.2918	22.7724	2.5972	1.5193	True
whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb	5	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	40.5021	26.1654	2.5972	14.3367	True
whisper_v3+fastconformer+w2vbert+hubert+cohere	5	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	24.157	23.1072	2.5972	1.0498	True
whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	5	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	39.523	24.5139	2.5972	15.0091	True
seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb	6	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	25.9996	23.6962	2.5972	2.3034	True
seamless_m4t+whisper_v3+w2vbert+hubert+wav2vec_sb+cohere	6	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	24.6427	22.9822	2.5972	1.6604	True
whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb+cohere	6	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2718	34.8691	24.7311	23.2406	2.5972	1.4904	True
seamless_m4t+whisper_v3+w2vbert+cohere	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2735	34.8706	26.3215	22.9131	2.5972	3.4084	True
seamless_m4t+whisper_v3+hubert+cohere	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2735	34.8706	23.885	23.0408	2.5972	0.8442	True
seamless_m4t+whisper_v3+wav2vec_sb+cohere	4	whisper_v3 whisper_v2	whisper_v3	whisper_v2	32.2735	34.8706	26.919	22.9337	2.5972	3.9853	True
seamless_m4t+fastconformer+w2vbert+hubert	4	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	41.4865	25.5054	0.3437	15.9811	False
seamless_m4t+w2vbert+hubert+wav2vec_sb	4	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	41.4916	41.1923	0.3437	0.2993	False
seamless_m4t+w2vbert+hubert+cohere	4	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	24.567	24.2306	0.3437	0.3364	False
seamless_m4t+w2vbert+wav2vec_sb+cohere	4	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	25.76	25.4952	0.3437	0.2649	False
seamless_m4t+hubert+wav2vec_sb+cohere	4	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	24.6145	24.2918	0.3437	0.3227	False
fastconformer+w2vbert+hubert+wav2vec_sb	4	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	64.7358	40.5021	0.3437	24.2336	False
fastconformer+w2vbert+hubert+cohere	4	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	39.0643	24.157	0.3437	14.9073	False
w2vbert+hubert+wav2vec_sb+cohere	4	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	39.5679	39.523	0.3437	0.045	False
seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert	5	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	26.0231	25.5054	0.3437	0.5176	False
seamless_m4t+fastconformer+w2vbert+hubert+wav2vec_sb	5	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	41.4916	25.9996	0.3437	15.492	False
seamless_m4t+fastconformer+w2vbert+hubert+cohere	5	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	24.567	21.4685	0.3437	3.0985	False
seamless_m4t+fastconformer+w2vbert+wav2vec_sb+cohere	5	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	25.76	21.4595	0.3437	4.3005	False
seamless_m4t+fastconformer+hubert+wav2vec_sb+cohere	5	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	24.6145	21.4867	0.3437	3.1279	False
seamless_m4t+w2vbert+hubert+wav2vec_sb+cohere	5	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	25.1784	24.6427	0.3437	0.5357	False
whisper_v3+fastconformer+w2vbert+hubert+cohere	5	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	24.3423	24.157	0.3437	0.1853	False

pool	k	pair	true_winner	pred_winner	wer_true_a	wer_true_b	wer_pseudo_a	wer_pseudo_b	gap_wer_true	gap_wer_pseudo	human_resolvable
fastconformer+w2vbert+hubert+wav2vec_sb+cohere	5	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	39.5679	24.7311	0.3437	14.8369	False
seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb	6	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	26.6374	25.9996	0.3437	0.6378	False
seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert+cohere	6	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	21.8521	21.4685	0.3437	0.3836	False
seamless_m4t+whisper_v3+fastconformer+w2vbert+wav2vec_sb+cohere	6	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	21.8689	21.4595	0.3437	0.4094	False
seamless_m4t+whisper_v3+fastconformer+hubert+wav2vec_sb+cohere	6	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	21.8224	21.4867	0.3437	0.3357	False
seamless_m4t+fastconformer+w2vbert+hubert+wav2vec_sb+cohere	6	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	25.1784	21.574	0.3437	3.6043	False
whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb+cohere	6	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	24.817	24.7311	0.3437	0.0859	False
seamless_m4t+whisper_v3+fastconformer+w2vbert+hubert+wav2vec_sb+cohere	7	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2718	21.9683	21.574	0.3437	0.3942	False
seamless_m4t+whisper_v3+fastconformer+cohere	4	fastconformer whisper_v3	fastconformer	whisper_v3	31.9297	32.2735	21.7397	21.3867	0.3437	0.3529	False
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	fastconformer whisper_v3	fastconformer	whisper_v3	31.9297	32.2718	27.676	27.4913	0.3421	0.1847	False
seamless_m4t+whisper_v3+fastconformer+hubert	4	fastconformer whisper_v3	fastconformer	whisper_v3	31.9297	32.2718	25.592	25.1931	0.3421	0.399	False
seamless_m4t+fastconformer+w2vbert+cohere	4	fastconformer whisper_v3	fastconformer	whisper_v3	31.9297	32.2718	26.6499	21.4151	0.3421	5.2349	False
seamless_m4t+fastconformer+hubert+cohere	4	fastconformer whisper_v3	fastconformer	whisper_v3	31.9297	32.2718	24.268	21.4903	0.3421	2.7778	False
seamless_m4t+fastconformer+wav2vec_sb+cohere	4	fastconformer whisper_v3	fastconformer	whisper_v3	31.9297	32.2718	27.2652	21.3904	0.3421	5.8748	False
whisper_v3+fastconformer+w2vbert+cohere	4	fastconformer whisper_v3	fastconformer	whisper_v3	31.9297	32.2718	25.1507	25.1236	0.3421	0.0271	False
whisper_v3+fastconformer+hubert+cohere	4	fastconformer whisper_v3	fastconformer	whisper_v3	31.9297	32.2718	24.0121	23.9625	0.3421	0.0496	False
seamless_m4t+whisper_v3+fastconformer+w2vbert+cohere	5	fastconformer whisper_v3	fastconformer	whisper_v3	31.9297	32.2718	21.7976	21.4151	0.3421	0.3826	False
seamless_m4t+whisper_v3+fastconformer+hubert+cohere	5	fastconformer whisper_v3	fastconformer	whisper_v3	31.9297	32.2718	21.9134	21.4903	0.3421	0.4232	False
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb+cohere	5	fastconformer whisper_v3	fastconformer	whisper_v3	31.9297	32.2718	21.7557	21.3904	0.3421	0.3653	False
seamless_m4t+fastconformer+w2vbert+wav2vec_sb	4	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2654	56.8546	26.6508	0.3373	30.2038	False
seamless_m4t+fastconformer+hubert+wav2vec_sb	4	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2654	44.2064	25.5286	0.3373	18.6778	False
whisper_v3+fastconformer+hubert+wav2vec_sb	4	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2654	42.5769	42.4835	0.3373	0.0934	False
fastconformer+w2vbert+wav2vec_sb+cohere	4	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2654	54.5584	24.8202	0.3373	29.7382	False
fastconformer+hubert+wav2vec_sb+cohere	4	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2654	41.6688	24.0835	0.3373	17.5853	False
seamless_m4t+whisper_v3+fastconformer+w2vbert+wav2vec_sb	5	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2654	27.1065	26.6508	0.3373	0.4557	False
seamless_m4t+whisper_v3+fastconformer+hubert+wav2vec_sb	5	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2654	25.9128	25.5286	0.3373	0.3842	False
whisper_v3+fastconformer+hubert+wav2vec_sb+cohere	5	fastconformer whisper_v3	fastconformer	whisper_v3	31.9281	32.2654	24.2109	24.0835	0.3373	0.1274	False
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	fastconformer whisper_v3	fastconformer	whisper_v3	31.9297	32.2654	28.7067	28.547	0.3357	0.1596	False

step7_floor_validated_ensemble_rdi

model	frac_zero_wer_samples	kendall_b_all	kendall_b_nonzero	n_nonzero	kendall_b_editdist
cohere	0.1408	0.3557	0.3026	2826	0.4773
seamless_m4t	0.1022	0.4641	0.411	2953	0.5893
fastconformer	0.0967	0.4851	0.4428	2971	0.6142
whisper_v3	0.0873	0.4791	0.43	3002	0.5967
whisper_v2	0.0721	0.4987	0.4559	3052	0.6165
hubert	0.0158	0.6754	0.6681	3237	0.8074

model	frac_zero_wer_samples	kendall_b_all	kendall_b_nonzero	n_nonzero	kendall_b_editdist
w2vbert	0.0103	0.7423	0.7374	3255	0.8677
xlsr	0.0012	0.7507	0.75	3285	0.8808
wav2vec_sb	0.0009	0.761	0.7605	3286	0.8829

step8_permutation_validated_ensemble_rdi

perm	order	n	identical	differing	identical_frac
0	cohere+wav2vec_sb+hubert+w2vbert+fastconformer+whisper_v3+seamless_m4t	3289	3289	0	1
1	fastconformer+hubert+w2vbert+seamless_m4t+whisper_v3+cohere+wav2vec_sb	3289	3289	0	1
2	hubert+cohere+w2vbert+wav2vec_sb+seamless_m4t+fastconformer+whisper_v3	3289	3289	0	1
3	cohere+seamless_m4t+hubert+fastconformer+w2vbert+whisper_v3+wav2vec_sb	3289	3289	0	1
4	wav2vec_sb+w2vbert+fastconformer+seamless_m4t+whisper_v3+cohere+hubert	3289	3289	0	1
5	cohere+wav2vec_sb+seamless_m4t+w2vbert+fastconformer+whisper_v3+hubert	3289	3289	0	1